

Differential Calculus — Chain Rule

Chapter 2: Sine of Linear Function | Page 33

PROBLEM STATEMENT

Trigonometric Function & Linear Angle

Find the derivative with respect to x of the function:

$$y = \sin(3x + 4)$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক ফাংশনটি উল্লেখ করি (Given Trigonometric Function)

$$y = \sin(3x + 4)$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\sin(3x + 4)]$$

STEP 3 সাইন ফাংশনের ওপর চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[\sin u] = \cos u \frac{du}{dx}$)

$$\frac{dy}{dx} = \cos(3x + 4) \cdot \frac{d}{dx}(3x + 4)$$

STEP 4 ভেতরের রৈখিক কোণের অন্তরক নির্ণয় করি (Differentiating Inner Linear Angle)

$$\frac{d}{dx}(3x + 4) = 3(1) + 0 = 3$$

STEP 5 মান বসিয়ে ধ্রুবক সহগকে সামনে লিখে চূড়ান্ত অন্তরজ পাই (Final Form)

$$\frac{dy}{dx} = \cos(3x + 4) \times 3 = 3 \cos(3x + 4)$$

FINAL ANSWER
$$\frac{dy}{dx} = 3 \cos(3x + 4)$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 04

✦ BOX 1: FORMULATION & LINEAR ANGLE RULES

- **Standard Trigonometric Formula:** $\frac{d}{dx} \sin(ax + b) = a \cos(ax + b)$.
- **Phase Shift Properties:** The constant term $+4$ represents a phase shift; its derivative is 0, leaving only the frequency/scaling factor $a = 3$.
- Thus, $\frac{d}{dx} \sin(3x + 4) = 3 \cos(3x + 4)$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing the Factor of 3:** Forgetting to differentiate the argument $(3x + 4)$ and erroneously writing $\frac{dy}{dx} = \cos(3x + 4)$.
- **Multiplying Inside the Angle:** Never multiply the exterior constant into the argument (i.e., $3 \cos(3x + 4) \neq \cos(9x + 12)$).

🌐 BOX 3: REAL DOMAIN & OSCILLATORY PROPERTIES

- **Domain & Range:** Defined on $\mathbb{R} = (-\infty, \infty)$. The derivative values are bounded in $[-3, 3]$.
- **Periodicity:** The fundamental period of both $y = \sin(3x + 4)$ and $y' = 3 \cos(3x + 4)$ is $T = \frac{2\pi}{3}$.
- **Higher Derivatives Pattern:** The n -th derivative is:

$$\frac{d^n y}{dx^n} = 3^n \sin\left(3x + 4 + \frac{n\pi}{2}\right)$$

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE DEMONSTRATION)

- Let $u = 3x + 4$, so $y = \sin u$.
- $\frac{dy}{du} = \cos u$.
- $\frac{du}{dx} = 3$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (\cos u) \cdot 3 = 3 \cos(3x + 4)$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct identification and application of the trigonometric chain rule.
- **1.0 Mark:** Correct derivative of inner linear function $\frac{d}{dx}(3x + 4) = 3$.
- **1.0 Mark:** Final structured representation $3 \cos(3x + 4)$.